

Methanomicrobium phylotype are the dominant methanogen phylotype in the Murrah buffaloes

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Dear Editor—This refers to the paper “16S rDNA directed PCR primers and detection of methanogens in the bovine rumen” published in March; 42(3), 2006 issue (Skillman *et al.* 2006).

We are studying the molecular diversity of methanogens in Murrah buffaloes and we performed the PCR-Cloning assay described by authors for the detection of methanogen diversity in this article. The assay was carried out using the Archaea specific primers (Arch f364/Arch r1386), as described by the authors. However, we have observed a distinctly different methanogen phylotypes from Murrah buffaloes as compared with that reported by authors in bovine rumen.

The authors have reported that most of clones shows sequence similarity with *Methanobrevibacter* spp. and *Methanosphaera stadtmanae*. In their study, *Methanobrevibacter* spp. and *Methanosphaera stadtmanae* appears to be predominant ruminal methanogens in their samples. However, when we carried out the experiments under the same conditions as described by the author we got *Methanomicrobium* phylotype as the dominant methanogen phylotype present in the bovine rumen in North India.

In the present study, 96 clones were studied and the sequence analysis revealed sixteen different phylotypes

whose sequences were similar to that of cultivated methanogens. Of 16 phylotypes determined, thirteen show similarity with *Methanomicrobium mobile* (similarity value ranges from 52% to 100%) and were 98.9–99.8% similar to *Methanomicrobium mobile* and the two phylotypes show similarity with *Methanocorpusculum bavaricum* and *Thermoplasma acidophilum*.

It might be because of change in the environmental conditions, feeding habits or genetic factors. Thus, our results confirm that *Methanomicrobium* phylotype is the dominant methanogen phylotype present in Murrah buffaloes in India.

P.P. Chaudhary
National Dairy Research Institute
– Dairy Cattle Nutrition Division,
Nutrition Biotechnology Lab,
Karnal, Haryana, India

Reference

- Skillman, L.C., Evans, P.N., Strömpl, C. and Joblin, K.N. (2006) 16S rDNA directed PCR primers and detection of methanogens in the bovine rumen. *Lett Appl Microbiol* **42**, 222–228.